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Here Come the Suns: Stripped Binary Populations in Early Galaxy Formation

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ABSTRACT

With the advent of the James Webb Space Telescope's (JWST) spectroscopy, we can measure the ultraviolet (UV) continuum and the nebular emission lines powered by ionizing radiation at redshifts up to $z \sim 14$. Interpreting these JWST observations at high redshifts requires understanding the spectral energy distribution (SED) of each galaxy, and their star formation history. Most models used to study galaxy formation derive the SED from single, non-interacting stars, neglecting the role of binary interactions. However, in close binaries, one star can lose its hydrogen-rich envelope to a companion, leaving behind a hot, stripped helium star. Without their hydrogen envelope, these stripped stars are extremely hot and emit hydrogen- (H) and helium II- (He II) ionizing photons long after the most massive stars have died. Without stripped stars, single-star population models can underestimate the He II-ionizing output of galaxies in which star formation has ceased. The He II emission is attributed to other sources like active galactic nuclei (AGN) or very massive stars. In this work, I will add stripped binary populations to a semi-analytical model (for galaxy formation) called `DELPHI`; I will include the spectra from stripped binary star populations in the star-formation history of every `DELPHI` galaxy to understand how much of the He II emission in early galaxies comes from stripped binaries.

1. INTRODUCTION

The James Webb Space Telescope (JWST) has revolutionized the field of high-redshift galaxy formation by offering groundbreaking sensitivity, spatial resolution, and wavelength coverage. It has enabled astronomers to identify hundreds of star-forming galaxies at redshifts up to $z \sim 14$, rattling the scientific community with two interesting observations: an unexpected abundance of super-early, luminous galaxies at the bright end ($M_{UV} \lesssim -21$) of the ultraviolet luminosity function (UV LF); and a growing number of sources that show hard ionizing spectra with strong He II emission. Both observations constrain the intrinsic stellar SED of early galaxies.

To model and predict early galaxy formation, one must model the growth of structure in the early Universe. Fortunately, the cosmological principle suggests that on large scales, the early Universe is homogeneous and isotropic, and the first bound structures are low-mass dark matter halos that collapse into overdense regions. Due to their shallow potential wells and low masses, even a small amount of star formation can provide the necessary feedback to expel all star-forming gas from these systems. To resume star formation, a low-mass halo must wait for gas to be accreted from the surrounding intergalactic medium (IGM) or to be deposited via mergers. Thus, it can be postulated that at any given time, there exists a limiting star formation efficiency such that the energy produced by newly formed stars is sufficient to expel all remaining gas, at least temporarily. Dayal et al. (2014) and Mauerhofer et al. (2025a) developed a semi-analytical model called `DELPHI` that embeds this simple principle. It tracks the accretion- and merger-driven, hierarchial assembly of dark matter halo, baryonic and black hole masses of high-redshift ($z > 5$) galaxies.

In most models like `DELPHI` , it is assumed that stellar populations consist of single stars. These co-eval stellar populations (also referred to as simple stellar populations) describe a group of single stars that formed together at the same time from the collapse of a single molecular cloud, sharing a uniform initial metallicity. The co-eval stellar populations that `DELPHI` uses come from `Starburst99` models. At each time step, single-star population-synthesis tables are used to evaluate the light of a stellar population of a given age and metallicity. Therefore, for a chosen initial mass function (IMF), that table fixes the 1500 Å luminosity and the production rates of H- ($E_\gamma > 13.6$ eV) and He II-ionizing ($E_\gamma > 54.4$ eV) photons. However, binary interactions via mass transfer open a new possibility.

For instance, a stripped star is a hot, helium-rich core left after a binary companion removed its hydrogen envelope. These stars emit many photons with $E_\gamma > 13.6$ eV (H-ionizing) and even $E_\gamma > 54.4$ eV (He II-ionizing).

With the inclusion of stripped stars, we expect the following changes in our understanding of early galaxies:

- Stripped stars emit most of their ionizing photons after massive single stars have died. Once a burst of $10^6 M_\odot$ is injected into stellar population models, stripped binaries do not appear until 10 Myr after the burst. Once they appear, they dominate the amount of ionizing photons output thereafter (Götberg et al. 2019). As a consequence, single-star models may underestimate the H- and He II-ionizing photons produced in galaxies where star formation stopped more than about 10 Myr ago.
- The amount of light at 1500 Å (the non-ionizing UV light) added by stripped stars is extremely diminutive, unless star formation stopped more than ~ 500 Myrs ago (Götberg et al. 2019). Thus, we expect the UV LF and UV LD predicted by `DELPHI` with co-eval single-star populations to remain the same. We are curious to see changes in ionization efficiency ξ_{ion}, if any at all.
- A nebular He II emission is one of the best indicators for the first, metal-free (Population III or Pop III) stars. Stripped stars in ordinary, metal-enriched populations also produce He II-ionizing photons. If Pop III stars and stripped star populations both produce He II, can we use their emission-line strengths to prevent metal-enriched populations (Population II or Pop II) with stripped binaries from masquerading as Pop III stars?

2. STUDYING STRIPPED BINARIES IN EARLY GALAXIES

This work aims to add the stripped binary population to DELPHI and determine how it changes the SED of early galaxies, from the UV to the He II-ionizing range. We can divide the SED into three separate regions:

- **UV non-ionizing:** $E_\gamma < 13.6\,eV$. We can understand how photons behave in this region by directly observing the UV luminosity function, the UV luminosity density and the UV slope β.
- **H-ionizing** $E_\gamma > 13.6\,eV$. These photons are absorbed by the gas and seen through hydrogen recombination lines (Lyα, Hα, Hβ). We can use this region to find the photon rate $Q(H)$ and $\xi_{ion} = Q(H)/L_{UV}$.
- **He II-ionizing** $E_\gamma > 54.4\,eV$. These photons ionize He^+. They are seen through He II λ1640 and λ4686. We can use this region to find the photon rate Q(He II) and the hardness $Q(HeII)/Q(H)$.

2.1. *SEDs of Stripped Stars*

Stripped stars should add only a very small fraction of ionizing photons to the UV window while massive stars of a co-eval stellar population are alive. Their contribution grows towards higher photon energies and with time once massive stars cease to exist. Their ionizing emission is harder than that of massive stars, and they dominate the ionizing output of populations about 10 Myrs after star formation stops.

In Figure 1, I use the data from `Starburst99` for single stars and from Götberg et al. (2019) for stripped stars to remake Figure 7 in Götberg et al. (2019). The figure shows the ionizing spectrum for a population formed in a single instantaneous burst of $10^6 M\odot$ at four metallicities. From Figure 1, it is evident that stripped stars harden (a harder spectrum means there are more high-energy photons relative to low-energy photons) the ionizing spectrum. The spectra that includes stripped stars ($11 - 100$ Myr) falls off more slowly in energy compared to the single-star spectra ($2 - 10$ Myr).

As seen in Figure 2, after a burst, around $150 - 450$ stripped stars exist between 10 Myr to 2 Gyr (Götberg et al. 2019). Since `DELPHI` follows each burst up to 890 Myr, stripped stars can contribute ionizing photons to `DELPHI` bursts, and can harden the SED of older bursts.

2.2. *Star Formation Histories From `DELPHI`*

`DELPHI` incorporates 600 merger trees into its framework. Each tree begins at a redshift $z = 4.5$; for each merger tree, I will rebuild its star formation history in 30 Myr steps, from $z = 4.5$ up to the first halos of the tree.

At each step, I will record: the stellar mass formed (M_*^{new}), the star formation rate ($SFR = M_*^{new}/30Myr$), the redshift (z), the metallicity of the gas (Z) formed by the new stars, the total stellar mass (M_*), the mass of dust present at the end of the step (M_d), and the UV luminosity at 1500Å(L_{UV}). From these parameters, I will compute the average metallicity of all the stars formed in the tree ($\langle Z\rangle = \frac{(\sum M_*^{new})Z}{\sum M_*^{new}}$). Note that M_*^{new}, z, Z and SFR help probe the intrinsic SED, while M_d and f_{esc}^{UV} are quantities that affect the observed spectral luminosity.

Then, I will convolve the star formation histories derived from `DELPHI` with the spectra of Figure 1 to understand the effect of stripped binaries on the SED of early galaxies.

2.3. *Finding Where He-II Ionizing Photons Come From*

In single-star Pop II models, photons above 54.4 eV are rare, so nebular He II in a galaxy with a Pop II population is usually attributed to sources like AGN, accreting

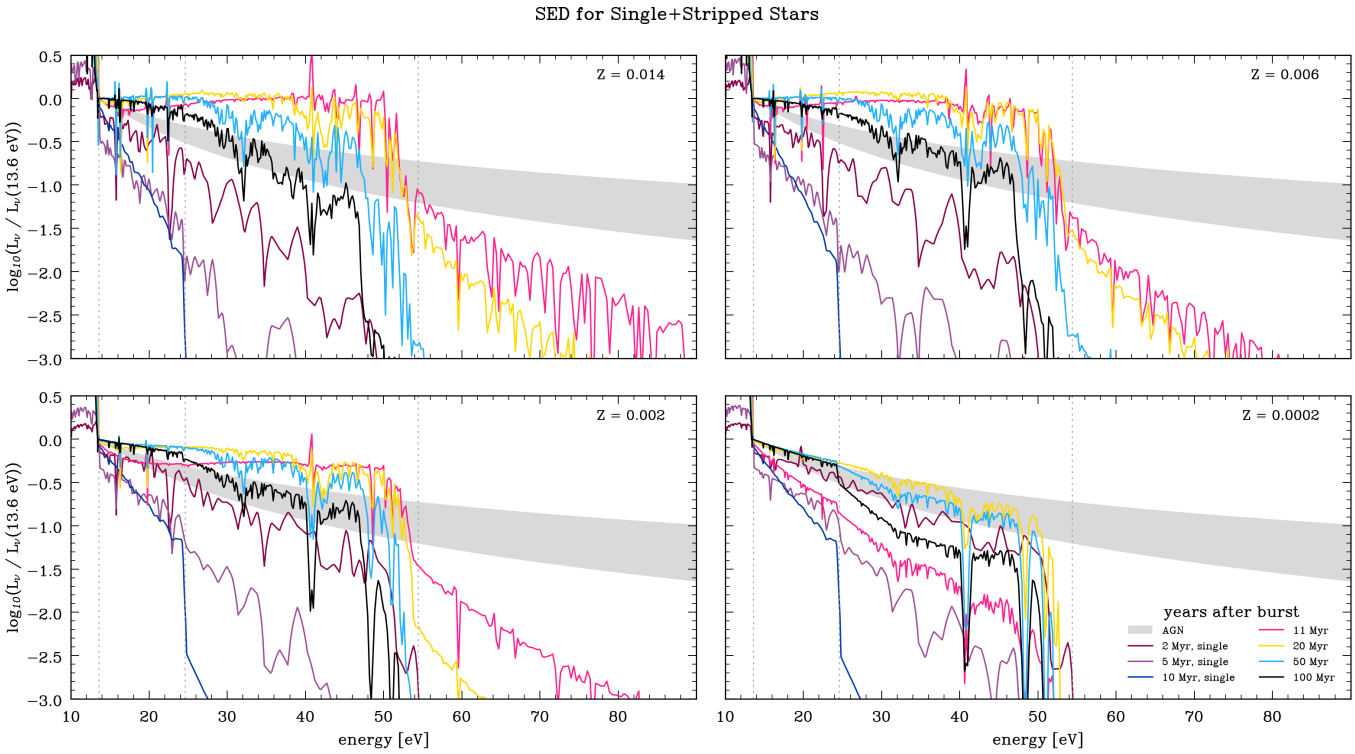


Figure 1. The ionization spectra of co-eval stellar populations for four metallicities: Z= 0.014, 0.006, 0.002, 0.02. They are made from the data published by Götberg et al. (2019). Single stars are shown 2, 5, and 10 Myr after the burst. Single+stripped stars are shown at 11, 20, 50 and 100 Myr. The gray band is accounting for the AGN with slopes $L_\nu \propto \nu^{-1.2} - \nu^{-2}$. The dotted, vertical lines are the ionization energies of H, He I and He II at 13.6, 24.6 and 54.4 eV respectively.

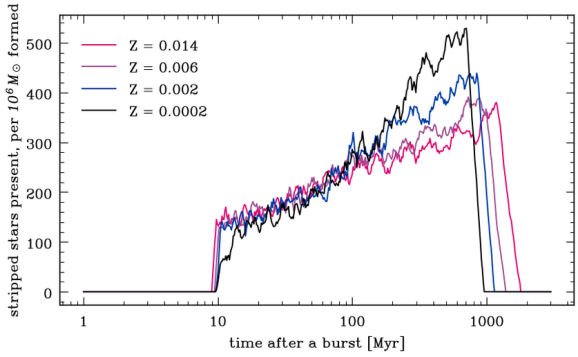


Figure 2. Number of stripped stars present in a stellar population at a given time after an instantaneous burst of $10^6 M\odot$ that forms stars. The binary population models are from Götberg et al. (2019). The figure is drawn for four metallicity values: Z = 0.014, 0.006, 0.002, 0.02.

black holes, X-ray binaries (Schaerer et al. 2019), or, at high redshift, Pop III stars. Even in metal-poor star-forming galaxies, He II is often stronger than single-star population models predict (Shirazi & Brinchmann 2012; Senchyna et al. 2017). Stripped binaries are missing from single-star models, yet they emit He II-ionizing photons (above 54.4 eV in Figure 1). An outcome of this work is to possibly answer whether some of the He II that is currently thought to originate from AGN, black holes or Pop III stars could actually be from stripped stars.

2.4. *Impact of Stripped Binaries on the Search for The First Stars*

Hard ionizing radiation, traced by nebular He II emission, is one of the best signatures of Pop III stars (Venditti et al. 2026). Figure 3 shows why this signature is difficult to use in practice (note that I did not produce this figure and it is Figure 10 in Venditti et al. (2026)). A Pop III-hosting galaxy passes through a pristine phase, a self-polluted phase (where it is pure Pop III), and a hybrid phase in which a Pop II population forms alongside the surviving Pop III stars. The Pop III-dominated phase ($M_{\star,PopIII} > 0.5 M_{\star,tot}$) lasts only $\sim$ 15-20 Myr.

In Figure 3, on the EW(He IIλ4686) versus [OIII]λ5007/$H\beta$ panel, the self-polluted and Pop III-rich galaxies lie above the lower limit of the Pop III threshold established in Nakajima & Maiolino (2022). Pop III-mid galaxies spread across the threshold, and Pop III-poor galaxies fall well below it, close to the region where AGN and other sources like black holes produce similar line ratios. Stripped helium stars are among these contaminants (Venditti et al. 2026), and they are one that I will model in `DELPHI`.

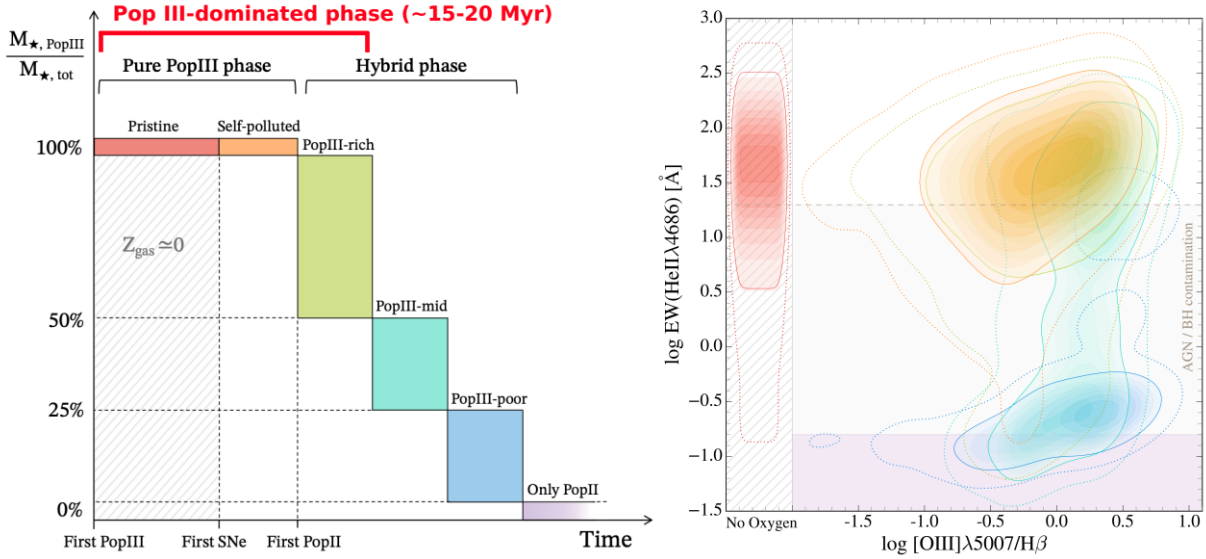


Figure 3. Fig. 10 of Venditti et al. (2026) (which was itself adapted from Rusta et al. (2025)). *Left:* evolutionary stages of a Pop III-hosting galaxy, ordered by the Pop III fraction of its stellar mass. The Pop III-dominated phase ($M_{\star,PopIII} > 0.5\, M_{\star,tot}$) lasts ∼15-20 Myr. *Right:* Distribution of Pop III-hosting galaxies predicted by the `NEFERTITI` model on the EW(He IIλ4686) versus [OIII] λ5007/Hβ diagram. Colors match the stages on the left; solid (dotted) contours enclose 68% (95%) of the galaxies. The dashed line marks the Pop III threshold of Nakajima & Maiolino (2022), and the lilac region below it the zone contaminated by AGN and direct-collapse black holes.

In a hybrid galaxy, the Pop II component contains binaries whose stripped stars keep emitting He II ionizing photons after its massive stars have died (Götberg et al. 2019). This can raise EW(He II) without any Pop III stars and push Pop III-poor or Pop II-only galaxies towards the Pop III threshold, creating false Pop III candidates.

Modeling stripped stars in `DELPHI` will yield the He II emission that Pop II populations alone can produce as a function of metallicity and redshift. This new boundary will help determine whether a He II detection is appropriate to strengthen the case for a Pop III star or not. A He II detection above the boundary is a Pop III star, while a detection below it is a Pop II contaminant masquerading as Pop III.

3. MILESTONES AND TIMELINE

I propose the following timeline between the months of October to May for my AST1501Y project, supervised by Prof. Pratika Dayal. In my timeline, I have taken into account the time required to learn `DELPHI`, meet 1501Y deadlines and conduct the research. A Gantt chart to summarize this timeline is shown in Figure 4.

- **October** - I have already learned DELPHI, compiled it, and run it with five seeds for the fiducial models on the 600 input merger trees. Using these runs, the UV LF, UV LD, halo mass function and stellar mass functions have been derived (Mauerhofer et al. 2025b).
- **November** - This period will be spent in writing and reviewing the 1501Y proposals and presentations. I will also begin adding the stripped-star spectra to the star formation histories of `DELPHI` galaxies.
- **December** - By the end of this month, the stripped binary spectra should be integrated into `DELPHI` completely. At the same time, the catalog from Hovis-Afflerbach et al. (2025) will be used to calculate the fraction of stripped-binary as a function of metallicity.
- **January** - I will start reviewing line strengths, finish the stripped-binary fraction plot against metallicity, decide which lines to model (e.g., $H\alpha, H\beta$) and how to treat escape and ionization parameters. The work progress will be shared as the midpoint for 1501Y.
- **February** - By quantifying the line strengths for each halo with and without stripped stars, my work will be compared against JWST results to confirm its validity.
- **March** - The project will be wrapped up by finishing the JWST line predictions. I will begin writing the final 1501 report.
- **April** - I will prepare and give the 1501 final presentation.

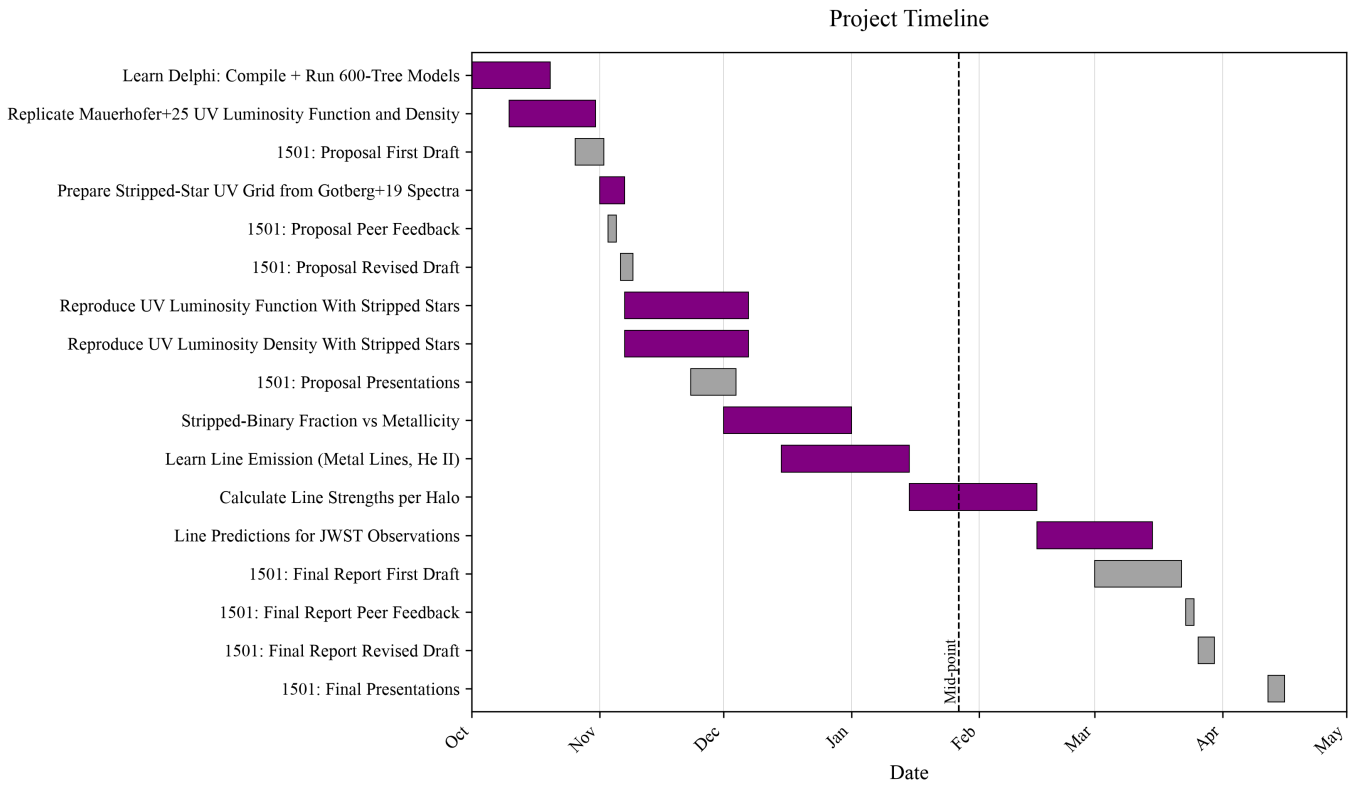


Figure 4. Gantt chart of the project timeline. In gray, I have the course deliverables. In purple, I have the research goals I intend to meet.

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Software: Astropy (Astropy Collaboration et al. 2013, 2018, 2022), `DELPHI` (Dayal et al. 2014; Mauerhofer et al. 2025a), Matplotlib (Hunter 2007), Pandas (Pandas 2020), Numpy (Harris et al. 2020),

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